

## ELECTIVE COURSE-6.4

| URA623: CONDITION MONITORING OF ROTATING MACHINERY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |     |
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| <p><b>Course Objective:</b> To utilize signal processing and data analysis tools with a focus on monitoring, assessing, predicting and diagnosing the condition of rotating machinery equipment. Students will follow a systematic process to conduct signal processing and data analysis. Students will be divided into groups to conduct term-project in applying this process for rotating machinery applications.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |     |
| <p><b>Syllabus</b></p> <p><b>Introduction:</b> Introduction of rotating machineries components and related fault symptoms e.g. Unbalance, Misalignment, Shaft cracks, Bearing faults, gear faults, Maintenance Principles and Condition Monitoring</p> <p><b>Introduction of MATLAB</b></p> <p><b>Measurement Signals:</b> Vibrations analysis, Acoustic emissions, Motor current signals, Different data measuring sensors e.g. Accelerometers, Eddy current probes, etc.</p> <p><b>Data Acquisition Principles:</b> Nyquist Theorem, Sampling frequency and number of samples, Windowing, Aliasing, Signal Conditioning and Filtering.</p> <p><b>Time domain analysis:</b> Statistical Parameters (Mean, median, RMS, Kurtosis, Crest Factor etc.), Gaussian distributions and their analysis, Time Synchronous Averaging, Case study including, auto-correlation, cross correlation.</p> <p><b>Frequency domain analysis:</b> Fourier Analysis, Fourier Transform, Fast Fourier Transform, Hilbert Transform, Envelope Analysis, Cepstrum Analysis</p> <p><b>Time frequency domain analysis:</b> Discrete Wavelet Transform, Continuous Wavelet Transform, Empirical Mode Decomposition, Hilbert Huang Transform</p> <p><b>Introduction to basic Machine Learning algorithms:</b> Artificial Neural Networks, Self-Organising Maps, Support vector Machines.</p> <p><b>Case Studies</b></p> <p><b>Research Assignment:</b> Students will be divided into groups to finish course projects. The course projects will consist of actual fault data from rotating machinery related topics. The course projects requires students to use analytical tools learned from this course to deal with real problems and which is also not limited to only use what you learned here. Innovative solutions and ideas are encouraged by bonus points. Students will have a final group presentation, where each group will demonstrate their understanding and use of the data analytics tools learned as applied to one real application.</p> |   |   |   |     |
| <p><b>Course Learning Outcomes (CLOs)</b></p> <p>The students will be able to:</p> <ol style="list-style-type: none"> <li>1. evaluate data sources and data quality in the context of rotating machinery faults</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |     |

*Approved in 109<sup>th</sup> meeting of the Senate held on March 16, 2023. Revised in 112<sup>th</sup> and 113<sup>th</sup> meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1<sup>st</sup> meeting of Academic Council held on September 11, 2025.*

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| <ol style="list-style-type: none"> <li>2. demonstrate proficiency at use of analytical tools and justify the use of methods selected</li> <li>3. use data and analytical tools to make predictive diagnosis regarding data from actual faults from rotating machinery equipment.</li> </ol>                   |
| <b>Text Books</b> <ol style="list-style-type: none"> <li>1 Randall, Robert Bond. Vibration-based condition monitoring: industrial, aerospace and automotive applications. John Wiley &amp; Sons, 2011.</li> <li>2 Bishop, Christopher M. Pattern recognition and machine learning. springer, 2006.</li> </ol> |

**Evaluation Scheme:**

| S.No. | Evaluation Elements                                      | Weightage (%) |
|-------|----------------------------------------------------------|---------------|
|       | MST                                                      | 30            |
|       | EST                                                      | 45            |
|       | Sessionals (Assignments/ Presentation, Technical Report) | 25            |

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